Complex Analysis Qualifying Exam Review

Fall 2026

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Contents

Chapter 1

Fall 2025

Problem 1.1. 5 Let $\{f_n\}$ be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that $f_n \to f$ uniformly on compact subsets of $\mathbb{D}$, where $f : \mathbb{D} \to \mathbb{C}$ is continuous. Prove that f is holomorphic on $\mathbb{D}$.

Proof. Morera's theorem. □

Problem 1.2. 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4+1}\,dx.$$

Proof. Use residue theorem for $z_1 = e^{\frac{\pi i}{4}}, z_2 = e^{\frac{3\pi i}{4}}$. □

Problem 1.3. 7 Let f be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all $z \in \mathbb{D}$. Show that

$$|f(z)| \geq 1 \qquad \text{for all } z \in \mathbb{D}.$$

Proof. Maximum modulus principle. □

Problem 1.4. 8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

Proof. Rouche's theorem. □

Chapter 2

Cauchy's integral formula

Problem 2.1. Evaluate the following integral:

$$\int_0^\infty \frac{1-\cos x}{x^2}dx$$

Proof. The answer is

$$\int_0^\infty \frac{1-\cos x}{x^2}dx = \frac{\pi}{2}$$

We use the contour of a semicircle skipping $x=0$. Let $f(z)=\frac{1-e^{iz}}{z^2}$, then by Cauchy's integral formula, we have

$$\int_{\gamma_R} f(z)dz + \int_{\gamma_\varepsilon} f(z)dz + \int_{|z|>\varepsilon} f(z)dz = 0 \tag{1}$$

where

$$\lim_{\varepsilon\to 0}\int_{|z|>\varepsilon} f(z)dz = 2\int_0^\infty \frac{1-\cos x}{x^2}dx$$

Hence it suffices to compute the first two terms in (1). Let $z=Re^{i\theta}$, we see that

$$\begin{aligned}\left|\int_{\gamma_R} f(z)dz\right| &= \left|\int_0^\pi \frac{1-e^{iRe^{i\theta}}}{R^2e^{2i\theta}}iRe^{i\theta}d\theta\right| \\ &\le \int_0^\pi \frac{2}{R}d\theta \to 0\end{aligned}$$

as $R\to\infty$. Similarly, we can compute $\int_{\gamma_\varepsilon} f(z)dz$ by setting $z=\varepsilon e^{i\theta}$ and letting $\varepsilon\to 0$.

$$\begin{aligned}\int_{\gamma_\varepsilon} f(z)dz &= \int_\pi^0 \frac{1-e^{i\varepsilon e^{i\theta}}}{\varepsilon^2 e^{2i\theta}}i\varepsilon e^{i\theta}d\theta \\ &= i\int_\pi^0 \frac{1-e^{i\varepsilon e^{i\theta}}}{\varepsilon e^{i\theta}}d\theta \to -\pi\end{aligned}$$

as $\varepsilon\to 0$, since

$$\lim_{a\to 0}\frac{1-e^{ia}}{a} = -i$$

□

Problem 2.2. Let U be a region. Suppose f is holomorphic in U, and a sequence $\{z_n\} \subset U$ converges to some $z \in U$. If $f(z_n) = 0$ for all n, show that $f \equiv 0$.

Proof. Since $z_n \to z$, we know $f(z) = \lim_{z_n \to z} f(z_n) = 0$. Assume f is not identically zero, then writing it as a power series, we have

$$f(w) = (w - z)^n g(w)$$

where $g(w) = a_n + a_{n+1}(w - z) + a_{n+2}(w - z)^2 + \dots$, for some $a_n \neq 0$. We know $g(z) = a_n \neq 0$, hence there exists a neighborhood V around z such that $g(w) \neq 0$ whenever $w \in V$. Hence, if $w \in V$, then $f(w) = 0$ iff $w = z$. This is a contradiction to $z_n \to z$ and $f(z_n) = 0$ for all n. $\square$

Warning 2.1. This shows that a nonzero holomorphic function has isolated zeros.

Problem 2.3 (Spring 2018, Q4). Let f be an entire function, suppose that for each z_0, the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient $a_n = 0$. Show that f must be a polynomial.

Proof. It suffices to show that for some $k \in \mathbb{N}$, $f^{(k)} \equiv 0$ because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some $k \in \mathbb{N}$ such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus $f^{(k)}$ has uncountably many zeros, which implies $f \equiv 0$.

Lemma 2.1. Let f be a nonzero holomorphic function, then f cannot have uncountably many zeros.

Proof. Let $Z = \{z \in \mathbb{C} : f(z) = 0\}$, we know that Z contains no limit point. We can tile the complex plane using unit squares and suppose Z is uncountable, the intersection of Z with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. $\square$

$\square$

Problem 2.4 (Fall 2019, Q2). Let f be an entire function such that

$$\sup_{|z| > R} |f(z)| \leq AR^k + B$$

for positive constants A, B and all R large. Then show f is a polynomial with degree at most k.

Proof. This problem is a direct application of Cauchy's inequality. Writing $f(z)$ as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any $m > k$, we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \to 0 \end{aligned}$$

as $R \to \infty$ because $m > k$. Hence we are done. □

Problem 2.5 (Fall 2023, Q1). (a) Show that if an entire function f satisfies

$$|f(z)| \leq \sqrt{|z|}$$

for all $z \in \mathbb{C}$, then $f(z)$ is a constant function.

(b) Why isn't the function $z^{1/2}$ a counterexample for Part (a)?

Proof. (a) We write f as the Taylor series about 0,

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

Using Cauchy's inequality, for any R large, if $n \geq 1$,

$$|f^{(n)}(0)| \leq \frac{n! \sup_{|z|=R} |f(z)|}{R^n} \to 0$$

as $R \to \infty$. Thus $f(z) = a_0$, for some $a_0 \in \mathbb{C}$.

(b) The function is not entire.

□

Problem 2.6 (Fall 2018, Q4). Suppose f, g are both entire functions such that $|f(z)| \leq |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f = cg$.

Proof. We will use Liouville's theorem. It suffices to show that $h = f/g$ continuously extends to an entire function. Suppose $g(z_0) = 0$ for some z_0, then we know $f(z_0) = 0$, hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where $s(z_0) \neq 0, t(z_0) \neq 0$. We must have $n \geq m$, suppose if not, i.e., $m > n$, then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting $z = z_0$. Thus z_0 is a removable singularity for $h = f/g$. This implies h can be extended to an entire function, as desired. $\square$

Chapter 3

Singularities

Problem 3.1. Compute
$$\int_0^\infty \frac{1}{1+x^2}dx$$

Problem 3.2. Compute the following integral: fix $0 < a < 1$,
$$\int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}dx$$

Proof. Use rectangles and the residue at πi. □

Problem 3.3 (Spring 2015, Q6). Suppose f is a nonconstant entire funcion, then show that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Proof. Suppose there exists $w \in \mathbb{C}$ such that for a small $r > 0$, $D_r(w) \cap f(\mathbb{C}) = \varnothing$. Then we consider the function
$$\frac{1}{f(z) - w}$$
This function is entire and bounded, implying f is constant, which is a contradiction. □

Problem 3.4. Assume that $f : \mathbb{C} \to \mathbb{C}$ is an entire function, not identically equal to 0, and let
$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$
Prove that if Z is unbounded, then f has an essential singularity at ∞.

Problem 3.5. Suppose f is an injective entire function, show f is linear.

Chapter 4

Conformal mapping

Problem 4.1 (Spring 2019, Q4)**.** Show that the punctured unit disk

$$\{z : 0 < |z| < 1\}$$

and the annulus

$$\{z : 1 < |z| < 2\}$$

cannot be conformally equivalent.

Problem 4.2 (Fall 2019, Q4)**.** Let f be a holomorphic function in the punctured disk

$$\{z : 0 < |z| < 2\}$$

satisfying

$$|f(z)| \leq \left(\log \frac{1}{|z|}\right)^{100} \qquad \text{in } \{z : |z| \leq 1/2\},$$

and

$$|f(z)| = 1 \qquad \text{on } |z| = 1.$$

(a) Show that f has a removable singularity at the origin.

(b) Show that if $f(z) \neq 0$ in $|z| < 1$, then f is constant.

Problem 4.3 (Fall 2019, Q4)**.** Let $f(z)$ be a holomorphic function on

$$D := \{z \in \mathbb{C} : |z| < 1\},$$

with $|f(z)| < 1$, and suppose $f(\alpha) = 0$ for some $|\alpha| < 1$. Show that for $z \in D$,

$$|f(z)| \leq \left|\frac{z - \alpha}{1 - \overline{\alpha} z}\right|.$$

Proof. Schwarz's lemma. □

Problem 4.4 (Spring 2016, Q5). Let $f : D \to D$ be a holomorphic function where

$$D = \{z \in \mathbb{C} : |z| < 1\}$$

is the unit disk. Prove that if f has at least 2 fixed points then f is the identity map.

Proof. Schwarz's lemma. $\square$

Chapter 5

Spring 2026

Problem 5.1. Let f be a holomorphic function on $\{z : |z| < 1\}$ that extends to a continuous function on $\{z : |z| \leq 1\}$ and maps the closed unit disk to itself. Furthermore, suppose

$$f(0) = f'(0) = \cdots = f^{(n-1)}(0) = 0.$$

Show that:

(1)
$$|f(z)| \leq |z|^n \qquad \text{for all } |z| \leq 1.$$

(2) If there is some $P \in D(0,1) \setminus \{0\}$ such that

$$|f(P)| = |P|^n$$

or if

$$|f^{(n)}(0)| = n!,$$

then

$$f(z) = e^{i\theta} z^n.$$

Problem 5.2 (Also Fall 2023, Q2)**.** Let $-1 < a < 0$. Compute

$$\int_0^\infty \frac{x^a}{1+x}\, dx.$$

Problem 5.3. Two parts.

(1) Let
$$f : U = \{z : 0 < |z| < 1\} \longrightarrow \mathbb{C}$$
be holomorphic and nonconstant. If there exists a sequence $\{z_n\} \subset U$ that converges to 0 and satisfies
$$f(z_n) = 0,$$
then prove that 0 is an essential singularity of f.

(2) Show that if f is an entire function that satisfies
$$\lim_{|z| \to \infty} f(z) = \infty,$$
then f is a polynomial.

Problem 5.4. Describe all injective holomorphic maps that map
$$\mathbb{C} \cup \{\infty\} \quad \text{to} \quad \mathbb{C} \cup \{\infty\}$$
and also map
$$\mathbb{R} \cup \{\infty\} \quad \text{to} \quad \mathbb{R} \cup \{\infty\}.$$